



HIGHER EDUCATION COMMISSION

H-9, ISLAMABAD, PAKISTAN, Website: <http://www.hec.gov.pk>

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Subject: National Skill Competency Test for IT Graduates (7th & 8th Semester Students of all HEIs)

Dear Vice Chancellor/Rector,

The Information Technology (IT) sector is experiencing rapid transformation, driven by emerging technologies, accelerated digitalization, and growing global integration. This dynamic landscape demands a workforce equipped with relevant, applied skills, presenting both challenges and opportunities for higher education institutions. Strengthening industry-academia collaboration through structured interventions is essential to enhance the employability, adaptability, and market readiness of IT graduates.

2. Pursuant to the Prime Minister's directives, communicated through the Pakistan Software Export Board (PSEB) vide letter No. PSEB/Skills/2025-553 dated April 25, 2025, the Higher Education Commission (HEC), in collaboration with the Ministry of Information Technology & Telecommunication (MoITT), PSEB, Pakistan Software Houses Association (P@SHA), and the National Computing Education Accreditation Council (NCEAC), has undertaken a structured consultative process to address the industry-academia skill gap. Thereafter, a series of consultative meetings were convened with Vice Chancellors and senior academic leadership of universities across Pakistan from November 05 to 12, 2025. The discussions centered on aligning computing education with industry needs, enhancing assessment credibility, and improving graduate employability. The recommendations from these consultations were subsequently presented to the Steering Committee to develop design of this initiative.
3. Following the decisions of the Steering Committee on the National Skill Competency Test for IT Graduates during its meetings from November 19 to December 24, 2025, the broader contours to conduct the National Skill Competency Test for IT Graduates are as under:
 - The Virtual University of Pakistan (VU) shall serve as the testing body for conducting the computer-based test. Coordination, verification, and facilitation shall be carried out through a Secretariat established at NCEAC, under the oversight of HEC, MoITT, PSEB, and P@SHA.
 - The test shall be administered on a biannual basis. The first sitting is scheduled for **April 04-05, 2026**. All students enrolled in the 7th and 8th semesters of Computer Science, Information Technology, Software Engineering, Artificial Intelligence, Data Science, and other relevant computing programs at HEC-recognized universities shall be eligible. To ensure nationwide coverage, credibility of results, and effective benchmarking, full participation of all eligible students from HEIs is strongly encouraged.
 - In case the universities/HEIs that do not participate in this exercise (i.e. fail to: nominate a Focal Person, facilitate student registration, and encourage eligible students to appear) will be placed below the lowest performance category in the national ranking and benchmarking of CS/IT programs. Similarly, the participating universities will be categorized (e.g., X, Y, Z) based on their students' overall performance in the assessment test, with higher categories reflecting stronger alignment with industry competencies and better graduate outcomes.
 - The test will be based on industry-endorsed competency areas (**Annexure-A**), finalized by a Technical Committee led by P@SHA, comprising representatives from industry, academia, accreditation bodies, and relevant stakeholders.



4. Participation is **free for all eligible students**. Qualifying students will:
- Receive a joint certificate issued under the aegis of HEC, PSEB, and P@SHA.
 - Be included in a national, skill-mapped talent pool accessible to local and international IT companies.
 - Be eligible for one (01) academic credit awarded by their respective institutions.
 - Gain enhanced access to industry opportunities such as internships, apprenticeships, employment pathways, and international certification programs.
5. The entire exercise is being undertaken with the objective of
- Credible data for national benchmarking and ranking of CS/IT programs.
 - Actionable insights for curriculum improvement and outcome-based education.
 - A structured pathway for future HEC funding and quality enhancement initiatives.
 - Access to a verified pool of IT graduates, strengthen industry participation in skills development, and support evidence-based workforce planning and policy formulation.
6. Foregoing in view, all HEIs are requested to disseminate this information widely among relevant students and faculty. Each university is required to nominate a Focal Person to liaise with the NCEAC Secretariat, led by Mr. Jawad Bukhari, Program Manager NCEAC (mbukhari@hec.gov.pk; 051-90402771; 03335298642). The Focal Person will facilitate in student registration, verification, and facilitation. The nomination details must be communicated to Mr. Jawad Bukhari by **February 06, 2026**.
7. HEC reiterates its commitment and support to undertaking meaningful measures for advancing a common agenda and initiatives.

(Prof Dr. Zia Ul Haq, S.I)

Distribution:

- Vice Chancellors/Rectors/Head of Public and Private Universities

Copy to:

- Chairman, HEC
- Members of the Steering Committee-
- Vice Chancellor, Virtual University
- Chairman, NCEAC
- CEO, PSEB
- Chairman, P@SHA
- Director General (Academics & Curriculum), HEC
- Director General (Quality Assurance), HEC



Areas of Competencies for National Skill Competency Test of IT Graduates

Areas of Competencies	Weightage %	Topics
Computer Networks and Cloud Computing	10%	1 - Data Communication
		2 - Computer Networks
		3 - Data Link Layer
		4 - Network Layer
		5 - Transport Layer
		6 - Application Layer
		7 - Wireless Networks
		8 - Cloud Computing
		9 - Network Security (Networks Perspective)
		10 - Next Generation Networks
Programming (C++/Java/Python)	10%	1. Programming Fundamentals
		2. Data Types & Variables
		3. Operators & Expressions
		4. Control Structures
		5. Functions / Methods,
		6. Input / Output Handling,
		7. Strings & Text Processing
		8. Arrays & Collections
		9. Object-Oriented Programming (OOP)
		10. Memory Management Concepts
		11. Exception & Error Handling
		12. Modules, Packages & Libraries
		13. Advanced Programming Concepts
		14. Concurrency & Parallelism (Introductory)
		15. Debugging, Testing & Optimization
		16. Software Development Practices
Data Structures & Algorithms	10%	1 - Foundations of Data Structure and Algorithms
		2 - Linear Data Structures
		3 - Non-Linear Data Structures
		4 - Searching Algorithms
		5 - Sorting Algorithms
		6 - Hashing
		7 - Tree Algorithms
		8 - Graph Algorithms
		9 - Algorithm Design Techniques
		10 - Advanced Data Structures
		11 - String Algorithms
		12 - Complexity & Optimization
Operating Systems	5%	1. Introduction to Operating Systems
		2. Operating System Structures
		3. Process Management
		4. CPU Scheduling
		5. Thread Management

		6. Concurrency & Synchronization
		7. Deadlocks
		8. Memory Management
		9. File System Management
		10. Secondary Storage Management
		11. Input / Output Systems
		12. Protection & Security
Software Engineering	10%	1. Introduction to Software Engineering
		2. Software Process Models
		3. Agile Software Development
		4. Software Requirements Engineering
		5. Software Project Management
		6. Software Design
		7. Software Architecture
		8. User Interface Design
		9. Software Implementation & Coding
		10. Software Testing
		11. Software Maintenance & Evolution
		12. Software Quality Assurance
		13. Software Metrics & Measurement
		14. Software Configuration Management
		15. Software Risk Management
		16. Software Security Engineering
Web Development	10%	1. Introduction to Web Development
		2. Web Architecture & Protocols
		3. HTML Fundamentals
		4. CSS Fundamentals
		5. Advanced CSS & Responsive Design
		6. JavaScript Fundamentals
		7. Advanced JavaScript
		8. Frontend Frameworks & Libraries
		9. Backend Development Fundamentals
		10. Server-Side Programming
		11. Databases for Web Applications
		12. Web Security
		13. Web Performance & Optimization
		14. Web Testing & Debugging
		15. Deployment & Hosting
		16. Web APIs & Integration
		17. Modern Web Development Practices
AI / Machine Learning and Data Analytics	10%	1. Introduction to AI, ML & Data Analytics
		2. Mathematical Foundations
		3. Python for AI & Data Analytics
		4. Data Collection & Pre-processing
		5. Exploratory Data Analysis (EDA)
		6. Supervised Learning

		7. Ensemble Learning
		8. Unsupervised Learning
		9. Model Evaluation & Validation
		10. Feature Engineering & Selection
		11. Deep Learning Fundamentals
		12. Advanced Deep Learning
		13. Natural Language Processing (NLP)
		14. Computer Vision
		15. Big Data Analytics (Introductory)
		16. Model Deployment & MLOps Basics
		17. AI Ethics, Security & Privacy
Cyber Security	5%	1. Introduction to Cyber Security
		2. Security Fundamentals & Principles
		3. Cryptography Basics
		4. Network Security
		5. Operating System Security
		6. Web Application Security
		7. Malware & Attack Techniques
		8. Authentication & Access Control
		9. Secure Software Development
		10. Wireless & Mobile Security
		11. Cloud & Virtualization Security
		12. Digital Forensics
		13. Incident Response & Management
		14. Security Monitoring & Auditing
		15. Cyber Laws & Ethics
		16. Emerging Trends in Cyber Security
Databases	10%	1. Introduction to Database Systems
		2. Database System Architecture
		3. Data Models
		4. Relational Database Concepts
		5. Relational Algebra & Calculus
		6. Structured Query Language (SQL)
		7. Advanced SQL
		8. Database Design & Normalization
		9. Transaction Management
		10. Concurrency Control
		11. Recovery Management
		12. Indexing & File Organization
		13. Query Processing & Optimization
		14. Database Security
		15. Distributed Databases
		16. NoSQL & Modern Databases
		17. Data Warehousing & Data Mining (Introductory)
	20%	1. Introduction to Problem Solving
		2. Problem Understanding & Analysis

Problem Solving And Anaytical Skills		3. Logical Reasoning Fundamentals
		4. Algorithms & Flow Control
		5. Data Representation & Abstraction
		6. Pattern Recognition & Generalization
		7. Mathematical & Quantitative Reasoning
		8. Algorithmic Thinking
		9. Critical Thinking & Decision Making
		10. Debugging & Error Analysis
		11. Complexity & Efficiency Awareness
		12. Problem Solving Using Programming
		13. Data-Driven Problem Solving
		14. Creative & Innovative Thinking
		15. Real-World Problem Solving
		16. Communication & Documentation of Solutions